

APPLICATION LOAD BALANCER

Class no:21

Application Load Balancer (ALB)

- **Definition:** An AWS service that distributes incoming application traffic across multiple targets (EC2, containers, IPs).
- **Protocol Used:** Works at **Layer 7 (Application Layer)** using **HTTP/HTTPS protocols**.
- **Intelligence:** It's "intelligent" because it can inspect the content of requests (headers, paths, hostnames) and make routing decisions based on rules.
- **Where Mostly Used:**
 - **Microservices architectures** → Different services (auth, catalog, payment) can be hosted separately and routed intelligently.
 - **Containerized applications** → Works seamlessly with ECS/EKS.

Path-Based Routing Example

- ALB can route traffic based on **URL path**:
 - `www.shop.com/auth/*` → Authentication service
 - `www.shop.com/catalog/*` → Product catalog service

This allows each microservice to scale independently and be managed separately.

Microservices vs Monolithic

Aspect	Monolithic Architecture	Microservices Architecture
Structure	Single, large codebase	Multiple small, independent services
Deployment	Entire app deployed together	Each service deployed independently
Scalability	Scale whole app	Scale specific services
Flexibility	Harder to adopt new tech	Each service can use different tech
Failure Impact	One bug can crash whole app	Failure isolated to one service

- **ALB** is crucial for modern **microservices** because it supports **path-based routing** and **content-based decisions** at the application layer.
- It uses **HTTP/HTTPS protocols** and provides intelligent traffic distribution.
- **Microservices** break applications into independent modules, unlike **monolithic** apps which are tightly coupled.

Hands on: Creating two instances for login and product, launching it using the ALB.

Step no: 1 Create an EC2 instance for login,

Name: **My-login-instance**

With OS type as Amazon Linux and with a key pair.

Launch an instance

Amazon EC2 allows you to create virtual machines, or instances, that run on the AWS Cloud. Quickly get started by following the simple steps below.

Name and tags

Name

My-login-instance

Add additional tags

Application and OS Images (Amazon Machine Image)

An AMI contains the operating system, application server, and applications for your instance. If you don't see a suitable AMI below, use the search field or choose [Browse more AMIs](#).

Search our full catalog including 1000s of application and OS images

Recents Quick Start

Amazon Linux

macOS

Ubuntu

Windows

Red Hat

SUSE Linux

Debian

[Browse more AMIs](#)
Including AMIs from AWS, Marketplace and the Community

Amazon Machine Image (AMI)

Amazon Linux 2023 kernel-6.1 AMI
ami-048f4445314bcaa09 (64-bit (x86), uefi-preferred) / ami-07efda20bfc979493 (64-bit (Arm), uefi)
Virtualization: hvm ENA enabled: true Root device type: ebs

Free tier eligible

Description

Amazon Linux 2023 (kernel-6.1) is a modern, general purpose Linux-based OS that comes with 5 years of long term support. It is optimized for AWS and designed to provide a secure, stable and high-performance execution environment to develop and run your cloud applications.

Amazon Linux 2023 AMI 2023.11.20260406.2 x86_64 HVM kernel-6.1

Architecture	Boot mode	AMI ID	Publish Date	Username
64-bit (x86)	uefi-preferred	ami-048f4445314bcaa09	2026-04-06	ec2-user

Verified provider

Instance type

Instance type

t3.micro

Family: t3 2 vCPU 1 GiB Memory Current generation: true On-Demand Linux base pricing: 0.0112 USD per Hour On-Demand SUSE base pricing: 0.0112 USD per Hour On-Demand Windows base pricing: 0.0204 USD per Hour On-Demand Ubuntu Pro base pricing: 0.0147 USD per Hour On-Demand RHEL base pricing: 0.04 USD per Hour

Free tier eligible

All generations

Compare instance types

Additional costs apply for AMIs with pre-installed software

Key pair (login)

You can use a key pair to securely connect to your instance. Ensure that you have access to the selected key pair before you launch the instance.

Key pair name - required

mumpk

Create new key pair

Network settings,

Choose: **Existing security group**

Choose: **default security group** (ensure http and https is enabled in the inbound rules)

▼ **Network settings** [Info](#)

Network | [Info](#)
vpc-0538c231a226fe3c9

Subnet | [Info](#)
No preference (Default subnet in any availability zone)

Auto-assign public IP | [Info](#)
Enable

Firewall (security groups) | [Info](#)
A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.

☐ Create security group ☒ Select existing security group

Common security groups | [Info](#)
Select security groups ▼ [Compare security group rules](#)

default sg-06d80e515c32b9515 ✕
VPC: vpc-0538c231a226fe3c9

Security groups that you add or remove here will be added to or removed from all your network interfaces.

Under **User data** → paste the comments given below

Note: for login mkdir /var/www/html/**login** and in echo "this is my home page"> /var/www/html/**login**/index.html

User data - optional | [Info](#)
Upload a file with your user data or enter it in the field.

[Choose file](#)

```
#!/bin/bash

#Update package repository and install Apache
sudo yum update -y
sudo yum install httpd -y

#Start the Apache and enable it to start on boot
sudo systemctl start httpd
sudo systemctl enable httpd

mkdir /var/www/html/login
echo "this is my home page"> /var/www/html/login/index.html

#Restart apache to apply changes
sudo systemctl restart httpd
```

☐ User data has already been base64 encoded

```
#!/bin/bash
```

```
#Update package repository and install Apache
```

```
sudo yum update -y
```

```
sudo yum install httpd -y
```

```
#Start the Apache and enable it to start on boot
```

```
sudo systemctl start httpd
```

```
sudo systemctl enable httpd
```

```
mkdir /var/www/html/login
```

```
echo "this is my home page">
/var/www/html/login/index.html
```

```
#Restart apache to apply changes
```

```
sudo systemctl restart httpd
```

Step no: 2 Create an EC2 instance for product,

Name: **My-product-instance**

With OS type as Amazon Linux and with a key pair.

Launch an instance [Info](#)

Amazon EC2 allows you to create virtual machines, or instances, that run on the AWS Cloud. Quickly get started by following the simple steps below.

Name and tags [Info](#)

[Add additional tags](#)

Application and OS Images (Amazon Machine Image) [Info](#)

An AMI contains the operating system, application server, and applications for your instance. If you don't see a suitable AMI below, use the search field or choose [Browse more AMIs](#).

Recents | **Quick Start**


Amazon Linux


macOS


Ubuntu


Windows


Red Hat


SUSE Linux


Debian

[Browse more AMIs](#)
Including AMIs from AWS, Marketplace and the Community

Amazon Machine Image (AMI)

Amazon Linux 2023 kernel-6.1 AMI
ami-048f4445314bcaa09 (64-bit (x86), uefi-preferred) / ami-07efda20bfc979483 (64-bit (Arm), uefi)
Virtualization: hvm ENA enabled: true Root device type: ebs

Description

Amazon Linux 2023 (kernel-6.1) is a modern, general purpose Linux-based OS that comes with 5 years of long term support. It is optimized for AWS and designed to provide a secure, stable and high-performance execution environment to develop and run your cloud applications.

Amazon Linux 2023 AMI 2023.11.20260406.2 x86_64 HVM kernel-6.1

Architecture	Boot mode	AMI ID	Publish Date	Username Info	Verified provider
64-bit (x86)	uefi-preferred	ami-048f4445314bcaa09	2025-04-06	ec2-user	

Instance type [Info](#) | [Get advice](#)

Instance type

t3.micro
Family: t3 2 vCPU 1 GB Memory Current generation: true On-Demand Linux base pricing: 0.0112 USD per Hour [Free tier eligible](#)
On-Demand SUSE base pricing: 0.0112 USD per Hour On-Demand Windows base pricing: 0.0204 USD per Hour
On-Demand Ubuntu Pro base pricing: 0.0147 USD per Hour On-Demand RHEL base pricing: 0.04 USD per Hour

☒ All generations [Compare instance types](#)

[Additional costs apply for AMIs with pre-installed software](#)

Key pair (login) [Info](#)

You can use a key pair to securely connect to your instance. Ensure that you have access to the selected key pair before you launch the instance.

Network settings,

Choose: **Existing security group**

Choose: **default security group** (ensure http and https is enabled in the inbound rules)

Network settings [Info](#) [Edit](#)

Network [Info](#)

vpc-0538c231a226fe3c9

Subnet [Info](#)

No preference (Default subnet in any availability zone)

Auto-assign public IP [Info](#)

Enable

Firewall (security groups) [Info](#)

A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.

☐ Create security group ☒ Select existing security group

Common security groups [Info](#)

Select security groups

default sg-06d80e515c32b9515 [X](#)

VPC: vpc-0538c231a226fe3c9

Security groups that you add or remove here will be added to or removed from all your network interfaces.

[Compare security group rules](#)

Under **User data** → paste the comments given below

Note: for login mkdir /var/www/html/**product** and in echo "this is my home page"> /var/www/html/**product**/index.html

User data - optional [Info](#)

Upload a file with your user data or enter it in the field.

[Choose file](#)

```
#!/bin/bash
```

```
#Update package repository and install Apache
```

```
sudo yum update -y
```

```
sudo yum install httpd -y
```

```
#Start the Apache and enable it to start on boot
```

```
sudo systemctl start httpd
```

```
sudo systemctl enable httpd
```

```
mkdir /var/www/html/product
```

```
echo "this is my product page"> /var/www/html/product/index.html
```

```
#Restart apache to apply changes
```

```
sudo systemctl restart httpd
```

☐ User data has already been base64 encoded

```
#!/bin/bash
```

```
#Update package repository and install Apache
```

```
sudo yum update -y
```

```
sudo yum install httpd -y
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```
#Start the Apache and enable it to start on boot
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```
sudo systemctl start httpd
```

```
sudo systemctl enable httpd
```

```
mkdir /var/www/html/product
```

```
echo "this is my home page">
```

```
/var/www/html/product/index.html
```

```
#Restart apache to apply changes
```

```
sudo systemctl restart httpd
```

Step no:3 Go to the default security group → edit inbound rules → add http and https → save rules.

Edit inbound rules [Info](#)

Inbound rules control the incoming traffic that's allowed to reach the instance.

Inbound rules [Info](#)

Security group rule ID	Type	Protocol	Port range	Source	Description - optional	
sgr-0ff6791321f532904	HTTPS	TCP	443	Custom	Q	Delete
sgr-06b9fe308e8427a99	All traffic	All	All	Custom	Q	Delete
sgr-08197f521eb29746a	SSH	TCP	22	Custom	Q	Delete
sgr-0f84d83f6745d9a22	HTTP	TCP	80	Custom	Q	Delete
sgr-02b0bb72fe6f5dc94	All TCP	TCP	0 - 65535	Custom	Q	Delete

[Add rule](#)

Rules with source of 0.0.0.0/0 or ::/0 allow all IP addresses to access your instance. We recommend setting security group rules to allow access from known IP addresses only.

[Cancel](#)

[Preview changes](#)

[Save rules](#)

Step no:4 Scroll down → go to target group → Create a target group for login instance.

Choose: **Instances**

Name: **My-login-tg**

Protocol: **HTTP**

Create target group

A target group can be made up of one or more targets. Your load balancer routes requests to the targets in a target group and performs health checks on the targets.

Settings - immutable

Choose a target type and the load balancer and listener will route traffic to your target. These settings can't be modified after target group creation.

Target type

Indicate what resource type you want to target. Only the selected resource type can be registered to this target group.

Instances

Supports load balancing to instances in a VPC. Integrate with Auto Scaling Groups or ECS services for automatic management.

Suitable for: ALB NLB GWLB

IP addresses

Supports load balancing to VPC and on-premises resources. Facilitates routing to IP addresses and network interfaces on the same instance. Supports IPv6 targets.

Suitable for: ALB NLB GWLB

Lambda function

Supports load balancing to a single Lambda function. ALB required as traffic source.

Suitable for: ALB

Application Load Balancer

Allows use of static IP addresses and PrivateLink with an Application Load Balancer. NLB required as traffic source.

Suitable for: NLB

Target group name

Name must be unique per Region per AWS account.

My-login-tg

Accepts: a-z, A-Z, 0-9, and hyphen (-). Can't begin or end with hyphen. 1-32 total characters; Count: 11/32

Protocol

Protocol for communication between the load balancer and targets.

HTTP

Port

Port number where targets receive traffic. Can be overridden for individual targets during registration.

80

1-65535

IP address type

Only targets with the indicated IP address type can be registered to this target group.

IPv4

Each instance has a default network interface (eth0) that is assigned the primary private IPv4 address. The instance's primary private IPv4 address is the one that will be applied to the target.

IPv6

Each instance you register must have an assigned primary IPv6 address. This is configured on the instance's default network interface (eth0). [Learn more](#)

VPC

Select the VPC with the instances that you want to include in the target group. Only VPCs that support the IP address type selected above are available in this list.

vpc-0538c231a226fe3c9
172.31.0.0/16

(default)



[Create VPC](#)

Protocol version

HTTP1

Send requests to targets using HTTP/1.1. Supported when the request protocol is HTTP/1.1 or HTTP/2.

HTTP2

Send requests to targets using HTTP/2. Supported when the request protocol is HTTP/2 or gRPC, but gRPC-specific features are not available.

gRPC

Send requests to targets using gRPC. Supported when the request protocol is gRPC.

Health checks → http → Health check path as **/login** (because this target group is for login)

Health checks

The associated load balancer periodically sends requests, per the settings below, to the registered targets to test their status.

Health check protocol

HTTP

Health check path

Use the default path of "/" to perform health checks on the root, or specify a custom path if preferred.

/login

Up to 1024 characters allowed.

Keep the rest as default → click Next.

▼ Advanced health check settings

Restore defaults

Health check port

The port the load balancer uses when performing health checks on targets. By default, the health check port is the same as the target group's traffic port. However, you can specify a different port as an override.

- ☒ Traffic port
☐ Override

Healthy threshold

The number of consecutive health checks successes required before considering an unhealthy target healthy.

2-10

Unhealthy threshold

The number of consecutive health check failures required before considering a target unhealthy.

2-10

Timeout

The amount of time, in seconds, during which no response means a failed health check.

seconds
2-120

Interval

The approximate amount of time between health checks of an individual target

seconds
5-300

Success codes

The HTTP codes to use when checking for a successful response from a target. You can specify multiple values (for example, "200,202") or a range of values (for example, "200-299").

Target optimizer - optional [Info](#)

Use a target control port when the target has a strict concurrency limit.

Target control port

The port on which the target communicates its capacity. This value can't be modified after target group creation.

Valid range: 1-65535

Attributes

[i](#) Certain default attributes will be applied to your target group. You can view and edit them after creating the target group.

► Tags - optional

Consider adding tags to your target group. Tags enable you to categorize your AWS resources so you can more easily manage them.

Cancel

Next

Choose include as pending below → add the login instance → next.

Register targets - recommended

This is an optional step to create a target group. However, to ensure that your load balancer routes traffic to this target group you must register your targets.

Available instances (1/2)

☐	Instance ID	Name	State	Security groups	Zone	Private IPv4 address	Subnet ID	Launch time
☐	i-058b9a7d05a37ddd4	My-Product-Instance	Running	default	ap-south-1a	172.31.40.47	subnet-01bf4d440c0912db3	April 8, 2026, 18:40 (UTC+05:30)
☒	i-01146ab917f123a9d	My-login-Instance	Running	default	ap-south-1a	172.31.34.166	subnet-01bf4d440c0912db3	April 8, 2026, 18:37 (UTC+05:30)

1 selected

Ports for the selected instances
Ports for routing traffic to the selected instances.

* Select a separate port for each instance

Include as pending below

1 selection is now pending below. Include more or register targets when ready.

Review targets

Targets (1)

☒ Show only pending

Instance ID	Name	Port	State	Security groups	Zone	Private IPv4 address	Subnet ID	Launch time
i-01146ab917f123a9d	My-login-Instance	80	Running	default	ap-south-1a	172.31.34.166	subnet-01bf4d440c0912db3	April 8, 2026, 18:37 (UTC+05:30)

1 pending

Cancel

Previous

Next

Click Create target group.

Step 1
Create target group
Step 2 - recommended
Register targets
Step 3
Review and create

Review and create

Review your target group configuration before creating

Step 1: Target group details

[Edit](#)

Name My-login-tg	Target type Instance	Protocol : Port HTTP: 80	Protocol version HTTP1
VPC vpc-0538c231a226fe3c9 ↗	IP address type IPv4		

Health check details

Health check protocol HTTP	Health check path /login	Health check port traffic-port	Interval 30 seconds
Timeout 5 seconds	Healthy threshold 5	Unhealthy threshold 2	Success codes 200

Step 2: Register targets

[Edit](#)

Targets (1)			
Instance ID	Name	Port	Zone
i-01146ab917f123a9d ↗	My-login-instance	80	ap-south-1a

[Cancel](#) [Previous](#) [Create target group](#)

Step no:5 Scroll down →go to target group → Create a target group for product instance.

Choose: **Instances**

Name: **My-Product-tg**

Protocol: **HTTP**

Create target group

A target group can be made up of one or more targets. Your load balancer routes requests to the targets in a target group and performs health checks on the targets.

Settings - immutable

Choose a target type and the load balancer and listener will route traffic to your target. These settings can't be modified after target group creation.*

Target type

Indicate what resource type you want to target. Only the selected resource type can be registered to this target group.

Instances

Supports load balancing to instances in a VPC. Integrate with Auto Scaling Groups or ECS services for automatic management.

Suitable for: [ALB](#) [NLB](#) [GWLB](#)

IP addresses

Supports load balancing to VPC and on-premises resources. Facilitates routing to IP addresses and network interfaces on the same instance. Supports IPv6 targets.

Suitable for: [ALB](#) [NLB](#) [GWLB](#)

Lambda function

Supports load balancing to a single Lambda function. ALB required as traffic source.

Suitable for: [ALB](#)

Application Load Balancer

Allows use of static IP addresses and PrivateLink with an Application Load Balancer. NLB required as traffic source.

Suitable for: [NLB](#)

Target group name

Name must be unique per Region per AWS account.

My-Product-tg

Accepts: a-z, A-Z, 0-9, and hyphen (-). Can't begin or end with hyphen. 1-32 total characters; Count: 13/32

Protocol

Protocol for communication between the load balancer and targets.

HTTP

Port

Port number where targets receive traffic. Can be overridden for individual targets during registration.

80

1-65535

IP address type

Only targets with the indicated IP address type can be registered to this target group.

IPv4

Each instance has a default network interface (eth0) that is assigned the primary private IPv4 address. The instance's primary private IPv4 address is the one that will be applied to the target.

IPv6

Each instance you register must have an assigned primary IPv6 address. This is configured on the instance's default network interface (eth0). [Learn more](#) [↗](#)

VPC

Select the VPC with the instances that you want to include in the target group. Only VPCs that support the IP address type selected above are available in this list.

vpc-0538c231a226fe3c9

172.31.0.0/16

(default) [↗](#)



[Create VPC](#) [↗](#)

Protocol version

HTTP1

Send requests to targets using HTTP/1.1. Supported when the request protocol is HTTP/1.1 or HTTP/2.

HTTP2

Send requests to targets using HTTP/2. Supported when the request protocol is HTTP/2 or gRPC, but gRPC-specific features are not available.

gRPC

Send requests to targets using gRPC. Supported when the request protocol is gRPC.

Health checks → http → Health check path as **/product** (because this target group is for product)

Health checks

The associated load balancer periodically sends requests, per the settings below, to the registered targets to test their status.

Health check protocol

HTTP

Health check path

Use the default path of "/" to perform health checks on the root, or specify a custom path if preferred.

/product

Up to 1024 characters allowed.

▼ **Advanced health check settings**

Restore defaults

Keep the rest as default → click **Next**.

▼ **Advanced health check settings**

Restore defaults

Health check port

The port the load balancer uses when performing health checks on targets. By default, the health check port is the same as the target group's traffic port. However, you can specify a different port as an override.

☒ Traffic port

☐ Override

Healthy threshold

The number of consecutive health checks successes required before considering an unhealthy target healthy.

5

2-10

Unhealthy threshold

The number of consecutive health check failures required before considering a target unhealthy.

2

2-10

Timeout

The amount of time, in seconds, during which no response means a failed health check.

5 seconds

2-120

Interval

The approximate amount of time between health checks of an individual target

30 seconds

5-300

Success codes

The HTTP codes to use when checking for a successful response from a target. You can specify multiple values (for example, "200,202") or a range of values (for example, "200-299").

200

Target optimizer - optional [Info](#)

Use a target control port when the target has a strict concurrency limit.

Target control port

The port on which the target communicates its capacity. This value can't be modified after target group creation.

Enter target control port

Valid range: 1-65535

Attributes

ⓘ Certain default attributes will be applied to your target group. You can view and edit them after creating the target group.

► **Tags - optional**

Consider adding tags to your target group. Tags enable you to categorize your AWS resources so you can more easily manage them.

Cancel **Next**

Choose **include as pending below** → add the product instance → next.

Register targets - recommended

This is an optional step to create a target group. However, to ensure that your load balancer routes traffic to this target group you must register your targets.

Available instances (2)

Instance ID	Name	State	Security groups	Zone	Private IPv4 address	Subnet ID	Launch time
i-058b9a7d03a77d5d4	My-Product-Instance	running	default	ap-south-1a	172.31.40.47	subnet-01f4d440c0912db5	April 8, 2025, 18:40 (UTC+05:30)
i-01146a09177123a9d	My-login-Instance	running	default	ap-south-1a	172.31.34.166	subnet-01f4d440c0912db5	April 8, 2025, 18:37 (UTC+05:30)

0 selected

Ports for the selected instances
Ports for routing traffic to the selected instances.

80

1 instance successfully matched ports with protocol

Include as pending below

1 selection is now pending below. Include more or register targets when ready.

Review targets

Targets (1)

Filter targets

Show only pending

Instance ID	Name	Port	State	Security groups	Zone	Private IPv4 address	Subnet ID	Launch time
i-058b9a7d03a77d5d4	My-Product-Instance	80	running	default	ap-south-1a	172.31.40.47	subnet-01f4d440c0912db5	April 8, 2025, 18:40 (UTC+05:30)

1 pending

Cancel Previous Next

Click Create target group.

- Step 1
- Create target group
- Step 2 - recommended
- Register targets
- Step 3
- Review and create

Review and create

Review your target group configuration before creating

Step 1: Target group details

Edit

Target group details

Name My-Product-tg	Target type Instance	Protocol : Port HTTP: 80	Protocol version HTTP1
VPC vpc-0538c231a226fe3c9	IP address type IPv4		

Health check details

Health check protocol HTTP	Health check path /product	Health check port traffic-port	Interval 30 seconds
Timeout 5 seconds	Healthy threshold 5	Unhealthy threshold 2	Success codes 200

Step 2: Register targets

Edit

Targets (1)

Instance ID	Name	Port	Zone
i-058b9a7d03a77d5d4	My-Product-Instance	80	ap-south-1a

Cancel Previous Create target group

Step no:6 Scroll down → Load balancer → Choose ALB.

Name: **My-ALB**

Scheme: **internet-facing**

Network mapping,

Choose: **default VPC**

Availability zone: Choose all the three

(the correct zone of the EC2 is enough, for more availability we are choosing all the three.)

Basic configuration

Load balancer name

Name must be unique within your AWS account and can't be changed after the load balancer is created.

My-ALB

A maximum of 32 alphanumeric characters including hyphens are allowed, but the name must not begin or end with a hyphen.

Scheme

Scheme can't be changed after the load balancer is created.

Internet-facing

- Serves internet-facing traffic.
- Has public IP addresses.
- DNS name resolves to public IPs.
- Requires a public subnet.

Internal

- Serves internal traffic.
- Has private IP addresses.
- DNS name resolves to private IPs.
- Compatible with the IPv4 and Dualstack IP address types.

Load balancer IP address type

Select the front-end IP address type to assign to the load balancer. The VPC and subnets mapped to this load balancer must include the selected IP address types. Public IPv4 addresses have an additional cost.

IPv4

Includes only IPv4 addresses.

Dualstack

Includes IPv4 and IPv6 addresses.

Dualstack without public IPv4

Includes a public IPv6 address, and private IPv4 and IPv6 addresses. Compatible with internet-facing load balancers only.

Network mapping

The load balancer routes traffic to targets in the selected subnets, and in accordance with your IP address settings.

VPC

The load balancer will exist and scale within the selected VPC. The selected VPC is also where the load balancer targets must be hosted unless routing to Lambda or on-premises targets, or if using VPC peering. To confirm the VPC for your targets, view [target groups](#).

vpc-0538c251a226fe3c9

(default)



Create VPC

IP pools

You can optionally choose to configure an IPAM pool as the preferred source for your load balancer's IP addresses. Create or view Pools in the [Amazon VPC IP Address Manager console](#).

Use IPAM pool for public IPv4 addresses

The IPAM pool you choose will be the preferred source of public IPv4 addresses. If the pool is depleted IPv4 addresses will be assigned by AWS.

Availability Zones and subnets

Select at least two Availability Zones and a subnet for each zone. A load balancer node will be placed in each selected zone and will automatically scale in response to traffic. The load balancer routes traffic to targets in the selected Availability Zones only.

ap-south-1a (aps1-az1)

Subnet

Only CIDR blocks corresponding to the load balancer IP address type are used. At least 8 available IP addresses are required for your load balancer to scale efficiently.

subnet-01bf4d440c0912db3

172.31.0.0/20

ap-south-1b (aps1-az3)

Subnet

Only CIDR blocks corresponding to the load balancer IP address type are used. At least 8 available IP addresses are required for your load balancer to scale efficiently.

subnet-0b3c21b6de4314d95

172.31.0.0/20

ap-south-1c (aps1-az2)

Subnet

Only CIDR blocks corresponding to the load balancer IP address type are used. At least 8 available IP addresses are required for your load balancer to scale efficiently.

subnet-0c305b80f72541592

172.31.16.0/20

Security group: default security group

Default action: Forward to target groups

Target group: My-login-tg → Add → My-Product-tg

Security groups

A security group is a set of firewall rules that control the traffic to your load balancer. Select an existing security group, or you can [create a new security group](#).

Security groups

Select up to 5 security groups



default (sg-06d80e515c32b9515)



Listeners and routing

A listener is a process that checks for connection requests using the port and protocol you configure. The rules that you define for a listener determine how the load balancer routes requests to its registered targets.

Listener HTTP:80

Remove

Protocol

HTTP

Port

80

1-65535

Default action

The default action is used if no other rules apply. Choose the default action for traffic on this listener.

Routing action

Forward to target groups

Redirect to URL

Return fixed response

Forward to target group

Choose a target group and specify routing weight or [create target group](#).

Target group

My-login-tg

Target type: Instance, IPv4 | Target stickiness: Off

HTTP



Weight

1

Percent

50%

Remove

My-Product-tg

Target type: Instance, IPv4 | Target stickiness: Off

HTTP



1

50%

Remove

0-999

+ Add target group

You can add up to 3 more target groups.

Target group stickiness

Enables the load balancer to bind a user's session to a specific target group. To use stickiness the client must support cookies. If you want to bind a user's session to a specific target, turn on the Target Group attribute Stickiness.

☐ Turn on target group stickiness

Listener tags - optional

Consider adding tags to your listener. Tags enable you to categorize your AWS resources so you can more easily manage them.

Add listener tag

You can add up to 50 more tags.

Add listener

You can add up to 40 more listeners.

Review the configurations → Create Load Balancer.

Review

Review the load balancer configurations and make changes if needed. After you finish reviewing the configurations, choose Create load balancer.

Summary

Review and confirm your configurations. [Estimate cost](#)

Basic configuration [Edit](#)

Name: My-ALB
Scheme: Internet-facing
IP address type: IPv4

Network mapping [Edit](#)

VPC: [vpc-0538c231a226fe3c9](#)
Public IPv4 IPAM pool: -
Availability Zones and subnets:

- ap-south-1a
[subnet-01bf4d440c0912db3](#)
- ap-south-1b
[subnet-0b3c21b6de4314d95](#)
- ap-south-1c
[subnet-0c305b80f72541592](#)

Security groups [Edit](#)

default:
[sg-06d80e515c32b9515](#)

Listeners and routing [Edit](#)

HTTP:80 | Forward to 2 target groups

Service integrations [Edit](#)

Amazon CloudFront + AWS Web Application Firewall (WAF): -
AWS WAF: -
AWS Global Accelerator: -

Tags [Edit](#)

-

Attributes

ⓘ Certain default attributes will be applied to your load balancer. You can view and edit them after creating the load balancer.

Creation workflow and status

► Server-side tasks and status

After completing and submitting the above steps, all server-side tasks and their statuses become available for monitoring.

[Cancel](#)

[Create load balancer](#)

Once created → Copy the **DNS of the Load balancer**.

Load balancer: My-ALB

[Details](#) | [Listeners and rules](#) | [Network mapping](#) | [Resource map](#) | [Security](#) | [Monitoring](#) | [Integrations](#) | [Attributes](#) | [Capacity](#) | [Tags](#)

Details

Load balancer type Application	Status Active	VPC vpc-0538c231a226fe3c9	Load balancer IP address type IPv4
Scheme Internet-facing	Hosted zone ZP978AFPLXTNKK	Availability Zones subnet-01bf4d440c0912db3 ap-south-1a (aps1-az1) subnet-0c305b80f72541592 ap-south-1c (aps1-az2) subnet-0b3c21b6de4314d95 ap-south-1b (aps1-az3)	Date created April 8, 2025, 18:51 (UTC+05:30)
Load balancer ARN arn:aws:elasticloadbalancing:ap-south-1:314383175192:loadbalancer/app/My-ALB/5d8065fb24e94395		DNS name info My-ALB-2029943255.ap-south-1.elb.amazonaws.com (A Record)	

Paste the **DNS** in a new Private Browser, we can see the output.

